



End (Odd) Semester Theory Examination- Nov-Dec 2025

Roll no.....

Name of the Program and semester: B.Tech CSE, III Sem.

Name of the Course: Python Programming

Course Code: TCS 346

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question are 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a. Create a Python module named stringutils.py containing the following functions:

count_words() – Returns the count of words in a given string.

reverse_string() – Returns the reversed version of the input string.

Import this module in another Python file and call all functions. Also implement exception handling to manage invalid inputs. (CO2)

- b. Explain the following types of Python function arguments with suitable examples:

- Positional Arguments
- Keyword Arguments
- Default Arguments
- Arbitrary Arguments (*args and **kwargs)

Also explain different forms of return statements in Python. (CO2)

- c. i) Differentiate between break and continue statements with appropriate examples.
ii) Write a Python program to read a paragraph from the user and perform the following operations:
- Count total consonants
 - Count uppercase and lowercase letters
 - Convert the entire paragraph into lowercase. (CO1)

Q2.

(2X10=20 Marks)

- a. Explain strings, lists and tuples in Python with appropriate examples. Compare them in terms of:

- Mutability
- Ordering
- Indexing
- Functions supported (Provide two built-in functions for each).

(CO3)

- b. Write a program to read a text file and:



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- (i) Identify the longest word and shortest word.
- (ii) Save both results into a new output file.

Also explain the role of `open()`, `read()`, `write()`, and `close()` in file handling. (CO3)

c. Write a Python program that:

- (i) Takes a list of integers from the user.
- (ii) Computes the square of each number using lambda function.
- (iii) Sorts the list in ascending order.
- (iv) Removes numbers less than 10.

(CO3)

Q3.

(2X10=20 Marks)

- a. Describe the concept of encapsulation in Python. Explain how public and private members are defined and accessed. Provide an appropriate Python example that demonstrates how private attributes can be accessed within a class. (CO4)
- b. Develop a program for a library system. Create a base class `Book` with attributes:
 - *Title*
 - *Author*
 - *price*

Include a method `calculate_total(qty)` to compute the total cost.

Create subclasses `EBook` and `PrintedBook` with the following conditions:

1. EBooks get a 10% discount.
2. Printed books include an additional ₹50 printing charge per unit.

Demonstrate polymorphism using `calculate_total()`. (CO4)

- c. A bank wants to create a program to manage different customer accounts using inheritance.

Base Class – *Account*

Attributes: *account_number*, *holder_name*, *balance*

Methods:

deposit(amount) – Add money to the account

withdraw(amount) – Deduct money if balance is sufficient

show_balance() – Display current account balance

Derived Class – *SavingsAccount*

Additional Attribute: *interest_rate*

Method:

add_interest() – Calculate interest using *interest_rate* and add it to the balance

Write a Python program that:

- (i) Creates an object of `SavingsAccount`.



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- (ii) Performs deposit and withdrawal operations.
- (iii) Adds interest using the method in SavingsAccount..
- (iv) Displays the updated balance after each operation.

(CO4)

Q4.

(2X10=20 Marks)

a. Using NumPy, develop a python program that:

- (i) Creates a 3×3 matrix
- (ii) Displays its data type, size, and number of dimensions
- (iii) Performs slicing on middle row
- (iv) Computes element-wise multiplication by 2
- (v) Flattens the array into a 1-D vector

(CO5)

b. Write Python program using Pandas to:

- (i) Create a DataFrame with StudentID, Course, and Marks for 5 students
- (ii) Find the total number of students enrolled.
- (iii) Find maximum marks per course.
- (iv) Join with another DataFrame containing StudentID and AttendancePercentage (CO5).

c. Predict output of the code snippets given below.

(i)

```
data = [3, 7]
data.insert(1, [9, 8])
data += [10, 11]
print(data)
```

(ii)

```
t = ("A", "B", "C", "D", "E")
print(t[0:3])
print(t[-4:-1])
print(t[::-1])
```

(iii)

```
import numpy as np
arr = np.array([2, 4, 6, 8, 10])
print(arr + 3)
print(arr[1:4])
```

(iv)

```
import pandas as pd
df = pd.DataFrame([1, 2, 3, 4])
f = lambda x: x**3
print(f(df))
```

(CO3, CO5)



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Q5.

(2X10=20 Marks)

- a. Define Exploratory Data Analysis (EDA). Discuss why EDA is important and describe its major steps in detail. Write a Python program using Seaborn to plot a histogram for the Age column of a sample dataset containing CustomerID, Age and AnnualIncome. (CO6)
- b. With the help of a diagram, explain the key parts of a boxplot. How does a boxplot help in identifying outliers? (CO6)
- c. Given a dataset with columns CustomerID, Gender, Age, Income, and PurchaseAmount. Write a Python program to:
 - (i) Replace missing Income values using median imputation.
 - (ii) Remove rows having missing values.
 - (iii) Compute correlation between Income and PurchaseAmount and interpret it.
 - (iv) Display all customers with PurchaseAmount > 500.

(CO6)